

Pinsara Weerasinghe

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Senior Backend Engineer | Java & AWS Cloud Microservices

University of Moratuwa Computer Science and Engineering Honors graduate with 6 years of experience architecting high-performance backend systems across the Fintech and Aviation domains. Spent 2.5 years in Fintech delivering low-latency Java and Spring Boot backend infrastructure to optimize transactional data processing. Advanced into Aviation tech for 3.5 years, designing cloud-native microservices and enterprise data hubs on AWS utilizing the Serverless Framework and AWS CDK. Combines deep server-side logic mastery with automated cloud provisioning to deliver fault-tolerant, high-throughput microservice architectures. Seeking to leverage scalable system design and cloud engineering expertise as a Senior Backend Engineer.

WORK EXPERIENCE

Serverless Guru • Remote • 02/2023 - Present

AWS cloud consultancy specializing in enterprise serverless architecture and microservices. Assigned as a Cloud Engineer to deliver high-scale data hub for Air Canada.

Senior Software Engineer / Cloud Consultant • Full-time

- **Architected a reusable Python utility library** for AWS Glue ETL jobs, optimizing data transformation workflows and **reducing AWS cloud compute costs by 20%**.
- **Engineered a cost-effective CloudWatch alarm triggering mechanism** utilizing log metric filters, streamlining automated alerting and **reducing CloudWatch expenditure by 2%**.
- **Engineered a real-time event persistence engine** using Kafka Connect and AWS Athena, enabling instant event debugging, rapid failure recovery, and **significantly enhancing overall system availability and fault tolerance**.
- **Optimized enterprise data migration pipelines**, refining execution logic and **reducing total data migration processing time by 30%**.
- **Led the development of a real-time customer data ingestion platform**, utilizing AWS Lambda and Apache Kafka topics to seamlessly process over **75% of total enterprise customer event workloads**.
- **Integrated OpenSearch indexing for high-traffic API endpoints**, offloading query patterns from primary databases to **reduce database load by 40%** and unblock downstream database integrations.

GTN Technologies (Previously Global Market Technologies) • Colombo, Sri Lanka • 06/2019 - 02/2023

A global fintech leader and institutional trading infrastructure provider connecting banks, asset managers, and wealth platforms to over 90 global capital markets via advanced e-trading engines.

Software Engineer • Full-time

- Engineered core real-time market data extraction components and OHLC layers to parse streaming feeds from stock exchanges.
- Designed data dissemination backend engines using **Java, Spring Boot, and WebSockets/TCP Sockets** for concurrent enterprise clients.
- Optimized text-search indexing speeds across database nodes by architecting wrappers around **Apache Solr** instances.
- Built asynchronous message queues using **Apache Kafka, AWS SQS, and AWS SNS** to handle background jobs.

Intern Software Engineer • Full-time

- Engineered automated log analysis and anomaly detection utilities for real-time market price backends while executing load/stress testing to optimize data throughput and platform stability under peak traffic spikes.

EDUCATION

Bachelor of Science (B.Sc. Hons) in Computer Science and Engineering

University of Moratuwa • Moratuwa, Sri Lanka • 09/2016 - 03/2021

* Completed intensive 4-year professional engineering curriculum specializing in advanced software architecture, data structures, cloud systems, and distributed computing databases.

G.C.E. Advanced Level Examination in (Physical Science Stream (Combined Mathematics, Physics, Chemistry))

Mo/Malwaththawala National School • Wellawaya, Sri Lanka • GPA: AAA • 01/2001 - 08/2014

* Achieved straight 'A' passes across all core technical subjects. Secured a highly competitive Z-Score of 2.3135, earning a District Rank of 2 out of thousands of candidates nationwide.

PROJECTS

Meridiano Ceylon – Tourism Booking Platform (www.meridianoceylon.com) • 04/2026 - 07/2026

Independent / Commercial Project

- **Technologies Used:** Next.js, Firebase, Tailwind CSS
- Architected and deployed a production Next.js and Firebase travel platform. Digitizing client operations with dynamic tour package management, real-time booking enquiries, secure authentication, and optimized cloud media storage.

Data Analytics & ML Platform • 05/2020 - 03/2021

University of Moratuwa (Final Year Research Project)

- **Technologies Used:** Apache Spark, NLP, Python, Machine Learning, Data Analytics
- Developed a final-year ML analytics framework using NLP and Apache Spark to process large-scale driver telemetry and feedback, computing real-time QoS scores and dynamic rankings.

MAS Holdings Quality Process Automation • 10/2020 - 12/2020

Fcode Labs

- **Technologies Used:** Flutter, Dart, Firebase
- Engineered a cross-platform Flutter/Dart app with Firebase integration to digitize garment evaluation workflows. Automating scheduling, measurement logging, and photo capture while developing a smart fashion tool to capture and visually compare clothing items.

VOLUNTEERING & LEADERSHIP

Department of Computer Science & Engineering, University of Moratuwa

Organizing Committee Member

Coordinated event logistics, technical showcases, and visitor engagement for Moratuwa's flagship CSE exhibition.

SKILLS

Cloud & Serverless Architecture: AWS (Lambda, CloudWatch, DynamoDB, Event-Driven Architecture, Glue, Microservices Architecture, OpenSearch, REST & GraphQL APIs, S3, Serverless Framework, SNS), SQS

Programming Languages: Dart, Java, JavaScript, Node.js, Python, SQL, TypeScript

Frameworks & Search Engines: Apache Solr, Express.js, Nest.js, Next.js, React, Spring Boot

Databases & Data Streaming: Amazon DynamoDB, Apache Kafka, Firebase, MySQL, PostgreSQL

DevOps, CI/CD & Tools: AWS CloudWatch Dashboards, Bamboo, Bitbucket, Docker, Git, Litestream

Personal Interests: Automotive Engineering & Mechanical Restorations, Open-Source Cloud Tooling, Travel & Outdoor Exploration